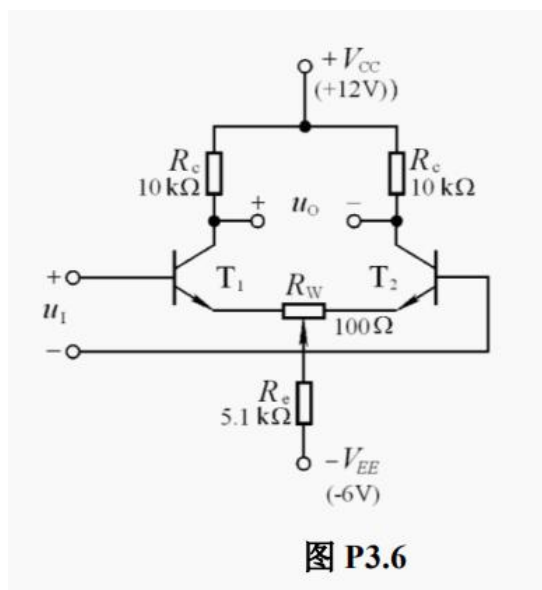


Homework 4

◇ Differential amplifier&Constant-current circuit: 3.6/3.8/4.3/4.13

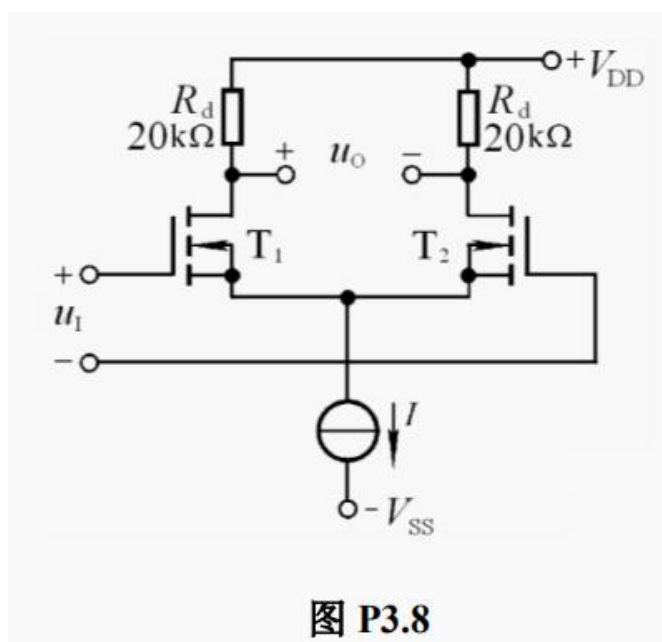
3.6 图 P3.6 所示电路参数理想对称, 晶体管的 β 均为 100, $r_{bb'}=100\ \Omega$, $U_{BEQ}=0.7$ 。试计算 R_w 滑动端在中点时 T_1 管和 T_2 管的发射极静态电流 I_{EQ} , 以及动态参数 A_d 和 R_i 。

3.6 The circuit parameters shown in figure P3.6 are ideal and symmetrical. The β of the transistors T_1 and T_2 are 100, $r_{bb'}=100\ \Omega$, and $U_{BEQ}=0.7\text{ V}$. The slider is shifted to the center of R_w . Please calculate the quiescent current I_{EQ} , AC voltage gain A_d and AC input resistance R_i .

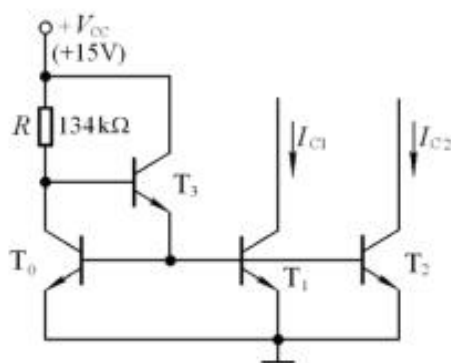


3.8 电路如图 P3.8 所示, T_1 和 T_2 的低频跨导 g_m 均为 10 mS 。试求解差模放大倍数和输入电阻。

3.8 A differential amplifier circuit is shown in figure P3.8, and the transconductance g_m of T_1 and T_2 is 10 mS . Please calculate the differential mode voltage gain and input resistance.



4.3 A multi-channel current source circuit is shown in figure **P4.3**. All transistors are the same and the U_{BE} is $0.7V$. Please find the value of I_{C1} and I_{C2} .



P4.3

- (1) 两个输入端中哪个是同相输入端，哪个是反相输入端；
- (2) T_3 与 T_4 的作用；
- (3) 电流源 I_3 的作用；
- (4) D_1 与 D_2 的作用。

- (1) For u_{i1} and u_{i2} , which is inverting input terminal and which is non-inverting input terminal;
- (2) The function of T_3 and T_4 ;
- (3) The function of current source I_3 ;
- (4) The function of D_1 and D_2 .

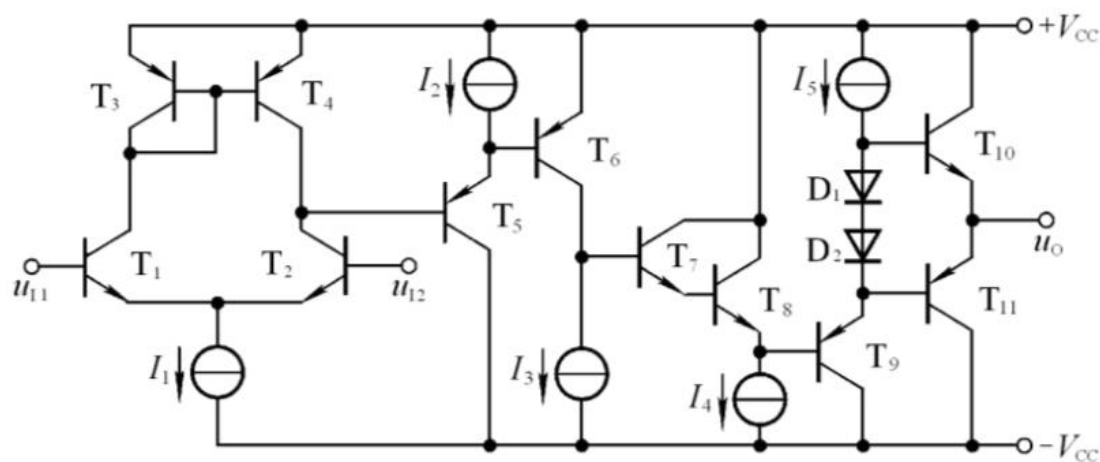


图 P4.13